

Jeongik Cho

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Summary

PhD in Computer Science with 4 SCIE-indexed journal publications and several international conference papers. Specialized in generative models (GANs, VAEs, Diffusion) and representation learning. Experienced in applying ML systems (PyTorch, TensorFlow, Hugging Face) to real-world tasks such as anomaly detection, clustering, and data generation. Currently extending expertise to LLMs.

Education

Concordia University, Montreal, Canada

PhD in Computer Science (Fast-Track)

Sep 2020 – Jul 2025

Konkuk University, Seoul, South Korea

BSc in Computer Science and Engineering

Mar 2016 – Mar 2020

Technical Skills

Frameworks/Tools: Python, PyTorch, TensorFlow, Hugging Face, Scikit-learn, SQL, Pandas

Expertise: Generative Models (VAEs, GANs, Diffusion), Representation Learning, Unsupervised Learning, Computer Vision, Anomaly Detection, Clustering, Regression, Large Language Models

Experiences

PhD Researcher, Concordia University, Montreal, Canada

Sep 2020 – Jul 2025

- Developed deep learning algorithms for generative models and representation learning.
- Applied methods to tasks such as unsupervised clustering, anomaly detection, and data manipulation, demonstrating the practical value of academic research.

Teaching Assistant, Concordia University, Montreal, Canada

Jan 2022 – Dec 2024

- Assisted in undergraduate courses by leading tutorial lectures and grading assignments, projects, and exams.

QA Game Tester, Altagram, Montreal, Canada

May 2023 – May 2024

Ongoing Projects

Adversarial Masked Diffusion for Path-Generation Decoupling - Proposed a diffusion-GAN hybrid framework that decouples path consistency from generation. [\[preprint\]](#)

Generative Modeling for LLMs - Exploring integration of generative modeling objectives into multi-token generation LLMs for flexible reasoning.

Publications

PhD thesis includes latest experiments of publications [\[pdf\]](#)

1. **Cho, J.** and Krzyzak, A., "Training Self-supervised Class-conditional GANs with Classifier Gradient Penalty and Dynamic Prior," under review. [\[link\]](#) [\[short ver\]](#) [\[code\]](#)
- We propose an algorithm to train a self-supervised class-conditional GAN using a classifier gradient penalty and a dynamic prior. This approach enables the model to represent data as a combination of continuous and discrete latent factors, which can be used for unsupervised clustering and classification tasks. **#Clustering #Classification #RepresentationLearning #GenerativeModel #ComputerVision**
2. **Cho, J.** and Krzyzak, A., "Efficient integration of perceptual Variational Autoencoder into Dynamic Latent Scale Generative Adversarial Network," Expert Systems, 2024. [\[paper link\]](#) [\[short ver\]](#) [\[code\]](#)
- We efficiently integrated a GAN with a perceptual VAE, demonstrating improved generative performance and enhanced inversion accuracy of the generative model. **#Representation Learning**

#GenerativeModel #ComputerVision

3. **Cho, J.** and Krzyzak, A., "Self-supervised Out-of-distribution Detection with Dynamic Latent Scale GAN," S+SSPR, 2022. [\[paper link\]](#) [\[short ver\]](#) [\[code\]](#)
 - We applied generative model inversion to map data into latent space and performed anomaly detection (out-of-distribution detection) using the log-likelihood scores of the inferred latent vectors. **#AnomalyDetection #GenerativeModel #ComputerVision**
4. **Cho, J.** and Krzyzak, A., "Dynamic Latent Scale for GAN Inversion," in ICPRAM, 2022. [\[paper link\]](#) [\[short ver\]](#) [\[code\]](#)
 - We proposed dynamically scaling the latent distribution to enhance the performance of GAN inversion and stabilize convergence. **#Representation Learning #GenerativeModel #ComputerVision**
5. **Cho, J.** and Yoon, K. "Conditional Activation GAN: Improved Auxiliary Classifier GAN," IEEE Access, 2020. [\[paper link\]](#) [\[code\]](#)
 - We proposed an algorithm that improves the performance of class-conditional GANs by leveraging multiple GAN loss functions while reducing the number of required hyperparameters. In addition, we introduced a method that ensures real and generated samples within each batch follow the same class distribution, enabling stable class-conditional generation even under class imbalance. **#Classification #GenerativeModel #ComputerVision**
6. Braun, A., Kohler, M., **Cho, J.**, and Krzyzak, A., "Analysis of the rate of convergence of two regression estimates defined by neural features," Electron. J. Stat., 2024. [\[paper link\]](#)
 - We proposed regression models training only output layers via regularized least squares, achieving theoretical convergence rates without backpropagation. **#NeuralNetwork #Regression**
7. Kohler, M., **Cho, J.**, and Krzyzak, A., "On the rate of convergence of an over-parametrized deep neural network regression estimate with ReLU activation function learned by gradient descent," Electron. J. Stat., 2025.
 - We proved that over-parameterized ReLU networks trained by gradient descent achieve dimension-free convergence under interaction model assumptions. **#NeuralNetwork #Regression**

Languages

Korean (native), English (proficient), Japanese (proficient, JLPT N1)